Mathematical notations usually express the idea more accurately and elegantly than words. In the study of line integrals, we need to integrate vector fields along a curve or a segment with specific endpoints. Thus, I invented the following notations:

- Segment from A to B: $[_A^B$, or even $[_{(a,b,c)}^{(x,y,z)}$ if we want to state the coordinates of the endpoints
- Arc from A to B: $(_A^B$, or even $(_{(a,b,c)}^{(x,y,z)}$.

These are very convenient. For example, it is easy to check that

$$\mathbf{F}(x,y) = (3x^2y^2 + 2xy^3, 2x^3y + 3x^2y^2)$$

is conservative. We know that a potential of a conservative vector field $\mathbf{F} = (M, N)$ is given by fixing A and integrating along any arc from A to P:

$$f(P) = \int_{(_A^P} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_A^P \mathbf{F} \cdot \mathrm{d}\mathbf{r}$$

or

$$f(x,y) = \int_{(a,b)}^{(x,y)} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_{(a,b)}^{(x,y)} M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y.$$

Thus, a potential of our $\mathbf{F}$ is given by integrating from $(0,0)$ to (x,y) along $[_{(0,0)}^{(x,0)}$ then $[_{(x,0)}^{(x,y)}$:

$$\begin{aligned} f(x,y) &= \int_{(0,0)}^{(x,y)} (3x^2y^2 + 2xy^3)\mathrm{d}x + (2x^3y + 3x^2y^2)\mathrm{d}y \\ &= \int_{[_{(0,0)}^{(x,0)}} (3x^2y^2 + 2xy^3)\mathrm{d}x + (2x^3y + 3x^2y^2)\mathrm{d}y \\ &\quad + \int_{[_{(x,0)}^{(x,y)}} (3x^2y^2 + 2xy^3)\mathrm{d}x + (2x^3y + 3x^2y^2)\mathrm{d}y \end{aligned}$$

$$
\begin{aligned}
&= \int_{\left[_{(0,0)}^{(x,0)}\right.} (3x^2y^2 + 2xy^3)\mathrm{d}x + \int_{\left[_{(x,0)}^{(x,y)}\right.} (2x^3y + 3x^2y^2)\mathrm{d}y \\
&= \int_0^x (3t^2 \cdot 0^2 + 2t \cdot 0^3)\mathrm{d}t + \int_0^y (2x^3t + 3x^2t^2)\mathrm{d}t \\
&= 0 + x^3y^2 + x^2y^3 = x^3y^2 + x^2y^3,
\end{aligned}
$$

where the second term of the first integral and the first term of the second integral after the second "=" are zero because along the segments $\left[_{(0,0)}^{(x,0)}\right.$ and $\left[_{(x,0)}^{(x,y)}\right.$ we have $\mathrm{d}y = 0$ and $\mathrm{d}x = 0$, respectively.

Of course this method applied to 3 dimensional vector fields

$$\mathbf{F}(x, y, z) = (M(x, y, z), N(x, y, z), H(x, y, z)).$$

Compare the above approach with the way Example 3 on §16.3 of the textbook is solved.

1 Minipage

The most beautiful formula is the Gauss-Bonnet formula

$$\chi(M) = \frac{1}{2\pi} \int_M K\mathrm{d}\sigma. \tag{1}$$

Geo Derivation.

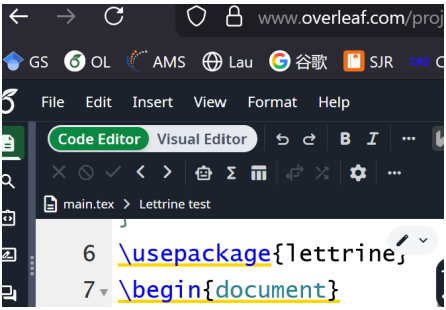
The most beautiful formula is the Gauss-Bonnet formula

$$\chi(M) = \frac{1}{2\pi} \int_M K\mathrm{d} \tag{2}$$

Geo Derivation.

2 Array

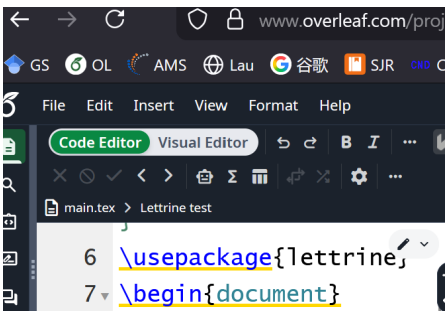
Find the volume of the region bounded above by the elliptical paraboloid $z = 10 + 2 + 3y^2$ and below by the rectangle.



At $x \in [_0^a$, to find mass of strip $\mathrm{d}Y$ above $[x, x+\mathrm{d}x]$, take $y \in [g_-(x), g_+(x)]$. Mass of $\mathrm{d}Q = [_x^{x+\mathrm{d}x} \times [_y^{y+\mathrm{d}y}$ is $\mathrm{d}m \approx f(x,y)\mathrm{d}x\mathrm{d}y$. View $\mathrm{d}Y$ as 1-D thin bar with 1-D density

$$\mathrm{d}m/\mathrm{d}y = f(x,y)\mathrm{d}x\,.$$

Find the volume of the region bounded above by the elliptical parabol $z = 10 + 2 + 3y^2$ and below by the rectangle.



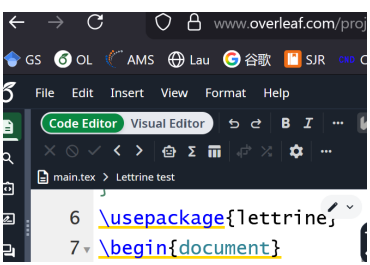
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$$\mathrm{d}m/\mathrm{d}y = f(x,y)\mathrm{d}x.$$

AAA

3 Lettrine test

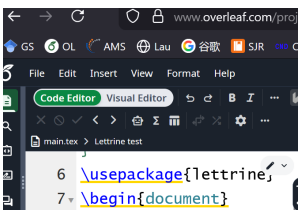
They are most closely related to the terns (family Sternidae), auks and skimmers, and more distantly to the waders. $\binom{1}{2}, \binom{0}{0}$



Are birds in the family Laridae. They are most closely related to the terns (family Sternidae), auks and skimmers, and more distantly to the waders.

$$A + B = C \tag{3}$$
$$= a + b \leq 0 \tag{4}$$

Most gulls belong to the large genus. Most gulls belong to the large genus Larus. are birds in the family Laridae. Most gulls belong to the large genus Larus. are birds in the family Laridae. They are most closely related to the terns (family Sternidae), auks and skimmers, and more distantly to the waders. Most gulls belong to the large genus Larus. are birds in the family Laridae. They are most closely related to the terns (family Sternidae), auks and skimmers, and more distantly to the waders. Most gulls belong to the large genus Larus. are birds in the family Laridae. They are most closely related to the terns (family Sternidae), auks and skimmers, and more distantly to the waders. Most gulls belong to the large genus Larus.



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